

Component Ablation — Point Forecast (Site 2, 15-min)

Model		Parameter	RMSE ↓	MAE ↓	R ² ↑	Delta R ² vs GRU	GRU Time/Epoch	Notes
GRU (Baseline)		239K	0.672	0.572	0.543	—	18s	2-layer, d=128
Mamba (Pure SSM)		120K	0.676	0.578	0.537	-0.6%	22s	d=64, ds=16, no LNN
Mamba + Spectral Loss		120K	0.673	0.575	0.540	-0.3%	25s	Frequency-domain aux loss
LNN-SSM (Ours)		148K	0.654	0.553	0.567	+2.4%	28s	GRU gate + Mamba SSM

96-step input -> 24-step output, MSE loss. Site 2 (200MW wind farm). LNN gating provides +2.4% R² with 28K fewer params than GRU.